

An Easy-to-Understand Explanation of the Research Paper

Paper: *An Analysis of Kalman Filter based Object Tracking Methods for Fast-Moving Tiny Objects*

This document explains **all important concepts** from the paper in a **step-by-step, intuitive way**, using **flowcharts, diagrams (ASCII), tables, and examples**, so that even if you're new to tracking/Kalman Filters, the logic becomes clear.

1. Big Picture: What Is This Paper About?

Problem Statement

Tracking **fast-moving tiny objects** (like a racquetball) is extremely hard because: - The object is **very small** → very few pixels - Moves **very fast** → motion blur - Changes direction **suddenly** → non-linear motion - Frequent **bounces & collisions**

Most modern trackers work well for **people, cars, bikes**, but **fail badly** for tiny fast objects.

What the Authors Did

- Created a **new racquetball dataset** (10,000 frames)
- Used **YOLOv8** to detect the ball
- Evaluated **5 Kalman Filter-based trackers**:
 - OCSORT
 - DeepOCSORT
 - ByteTrack
 - BoTSORT
 - StrongSORT

Core Goal

Check whether Kalman Filter-based trackers are suitable for fast-moving tiny objects

2. Object Tracking Basics (Foundation)

What Is Object Tracking?

Tracking =

"Detect object in frame 1 → follow the SAME object in frame 2, 3, 4..."

Two Main Tracking Approaches

1 Tracking-by-Detection (Used in this paper)

Frame → Object Detector → Bounding Box
↓
Tracker → ID + trajectory

- Detect object **every frame**
- Tracker **associates detections across frames**

Examples: - SORT - DeepSORT - ByteTrack

2 Tracking-without-Detection

Initial position → Predict future positions only

Examples: - Optical Flow - Mean Shift

✗ Not good for tiny objects (not enough appearance info)

3. Why Tiny Fast Objects Are So Difficult

Challenge	Why it breaks tracking
Tiny size	Very few pixels → weak features
High speed	Motion blur
Sudden direction change	Violates linear motion assumption
Occlusion	Detector misses object
Background clutter	False detections

⚠ Kalman Filter assumes smooth motion — racquetball motion is chaotic

4. Dataset Used in the Paper

Dataset Details

- 10,000 frames
- Resolution: 720p & 1280p
- Captured using ZED 2i stereo camera

- Only RGB frames used
- Manually annotated ball positions

Dataset Split

70% → Training
20% → Validation
10% → Testing

5. Object Detection System (YOLO)

Before tracking, the ball must be detected.

YOLO Models Compared

Model	Accuracy	Speed
YOLOv5	Good	Slow
YOLOv7	Worse	Slow
YOLOv8	Best balance	Fastest

✓ **YOLOv8 chosen** because: - Best for **small objects** - Much faster inference

Input Frame
↓
YOLOv8
↓
Ball Bounding Box

6. Evaluation Metrics (VERY IMPORTANT)

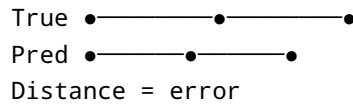
Traditional tracking metrics (MOTA, IDF1) **don't work well for single tiny objects**.

Metric 1: ADE (Average Displacement Error)

"How far is the predicted position from the true position?"

ADE = average Euclidean distance

Graphically:



True •————•————•
Pred •————•————•
Distance = error

Lower ADE = better tracking

Metric 2: AMD (Average Mahalanobis Distance)

"Error considering uncertainty and correlation"

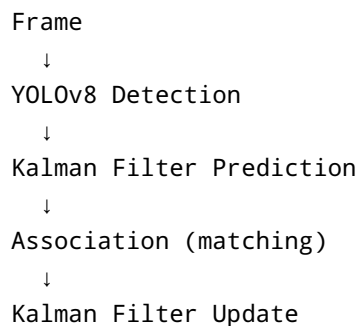
Why AMD? - Kalman Filter maintains **covariance** (uncertainty) - Mahalanobis distance uses that covariance

$$\text{AMD} = \sqrt{((x_{\text{pred}} - x_{\text{true}})^T \Sigma^{-1} (x_{\text{pred}} - x_{\text{true}}))}$$

Lower AMD = more confident + accurate tracking

7. Trackers Evaluated (High-Level Overview)

All trackers follow this **common pipeline**:



```
Frame
↓
YOLOv8 Detection
↓
Kalman Filter Prediction
↓
Association (matching)
↓
Kalman Filter Update
```

Difference lies in **how association & updates are done**.

8. Kalman Filter (Heart of the Paper)

What Kalman Filter Does

"Predict where the object will be, then correct using measurement"

Kalman Filter Flowchart

```
graph TD; A[Previous State] --> B[Prediction (motion model)]; B --> C[Predicted State]; C --> D[Measurement (YOLO box)]; D --> E[Correction (Kalman Gain)]; E --> F[Updated State];
```

State Representation (for tracking)

State $x = [x, y, vx, vy]$

- Position (x, y) - Velocity (vx, vy)

Constant Velocity Assumption ❌

Assumes:
Next position = current position + velocity × time



Racquetball breaks this assumption: - Sudden bounces - Direction flips

9. Tracker-by-Tracker Explanation

9.1 SORT (Base Tracker)

Detection → KF Prediction → IOU Matching → KF Update

Features: - No appearance features - Very fast - Very weak for tiny objects

9.2 ByteTrack (Fastest)

Key Idea:

"Don't discard low-confidence detections"

High-score detections → match first
Low-score detections → match later

✓ Very fast ✗ Very inaccurate for racquetball

Why? - Prioritizes speed - Weak correction when motion is erratic

9.3 BoT-SORT

Improvements: - Better Kalman state (includes width & height) - Camera Motion Compensation - IoU + Appearance fusion

Detection
↓
Camera Motion Compensation
↓
KF Prediction
↓
Association

Still fails due to unpredictable ball motion

9.4 OC-SORT (Observation-Centric)

Key Innovation:

"Trust observations more than prediction"

Techniques: - ORU (Re-update after occlusion) - OCM (Momentum-based direction) - OCR (Recovery step)

Better than SORT, but still limited

9.5 Deep OC-SORT (BEST PERFORMER)

Adds: - Deep appearance features - Adaptive weighting - Camera motion compensation

Motion Cost + Appearance Cost



Adaptive Fusion

✓ Lowest ADE & AMD ✗ Still not accurate enough for real use

10. Experimental Results (Key Findings)

Accuracy (Lower is better)

Tracker	ADE (avg)
DeepOCSORT	31 px ✓
OCSORT	79 px
ByteTrack	114 px ✗

Speed

Tracker	Inference Time
ByteTrack	Fastest
DeepOCSORT	Slightly slower

⚠ Trade-off: - Speed ↑ → Accuracy ↓

11. Why Kalman Filters Fail Here (Core Insight)

Kalman Filter Assumptions: - Linear motion - Smooth velocity - Gaussian noise

Racquetball Reality: - Non-linear - Sudden acceleration - Wall collisions

✗ Assumptions break → tracking drifts

12. Future Research Directions

The paper suggests:

1 Physics-based Tracking

Use ball dynamics:

- Bounce angle
- Energy loss
- Collision physics

2 Hybrid Models

- Learning + physics
- Detection + trajectory prediction

3 Specialized Tiny-Object Trackers

Not human-based trackers

13. Final Conclusion (In Simple Words)

Kalman Filter-based trackers are NOT suitable for fast-moving tiny objects like racquetballs

Even the best tracker: - Has 3–4× more error than normal benchmarks

 New tracking paradigms are needed.

If you want next: -  Viva-ready explanation -  PPT slides -  Handwritten-style notes -  Mathematical derivation simplification

Just tell me 